Hyperspectral Image Classification Using Attention-Based Bidirectional Long Short-Term Memory Network

Shaohui Mei, *Senior Member, IEEE*, Xingang Li, Xiao Liu, Huimin Cai, and Qian Du, *Fellow, IEEE*

***Abstract*— Deep neural networks have been widely applied to hyperspectral image (HSI) classification areas, in which recurrent neural network (RNN) is one of the most typical networks. Most of the existing RNN-based classifiers treat the spectral signature of pixels as an ordered sequence, in which only unidirectional correlation along the wavelength direction of adjacent bands is considered. However, each band image is related to not only its preceding band images but also its successive band images. In order to fully explore such bidirectional spectral correlation within an HSI, in this article, a bidirectional long short-term memory (Bi-LSTM)-based network is designed for HSI classification. Moreover, a spatial–spectral attention mechanism is designed and implemented in the proposed Bi-LSTM network to emphasize the effective information and reduce the redundant information among spatial–spectral context of pixels, by which the performance of classification can be greatly improved. Experimental results over three benchmark HSIs, i.e., Salinas Valley, Pavia Centre, and Pavia University, demonstrate that our proposed Bi-LSTM obviously outperforms several state-of-the-art unidirectional RNN-based classification algorithms. Moreover, the proposed spatial–spectral attention mechanism can further improve the classification accuracy of our proposed Bi-LSTM algorithm by effectively weighting spatial and spectral context of pixels. The source code of the proposed Bi-LSTM algorithm is available at https://github.com/MeiShaohui/Attention-based-Bidirectional-LSTM-Network.**

***Index Terms*— Attention network, classification, deep learning, hyperspectral image (HSI), recurrent neural network (RNN).**

I. Introduction

HYPERSPECTRAL imaging sensors can obtain abundant spectral information of objects while preserving their spatial information, which enables to explore spectral and spatial characteristics. Compared with convolutional color images or multispectral remote sensing images, hyperspectral images (HSIs) have greatly improved in information richness, leading to great interest in many applicational fields, such as medicine, agriculture, industry, and food [1]–[4].

HSI classification, which assigns labels to different pixels by exploring their spectral signature and spatial context, has attracted great attention in past decades. A simple way for such a purpose is directly feeding spectral pixel vectors into conventional classifiers [5]. For example, Melgani and Bruzzone [6] and Camps-Valls *et al.* [7] addressed the problem of the classification of HSIs by support vector machines (SVMs). Ham *et al.* [8] and Belgiu and Drăguţ [9] proposed to classify HSIs using a random forest (RF) classifier. However, due to high spectral dimensionality, directly using the spectral information of HSIs can easily lead to the curve of dimensionality (i.e., the Hughes effect) [10], [11]. Therefore, many methods were proposed to explore discriminative features implied in the high-dimensional spectral signatures [12], [13], among which the representative algorithms are principal component analysis (PCA) [14]–[16], linear discriminant analysis (LDA) [17], manifold learning-based methods [18], [19], and graph embedding [20], [21].

In recent years, many attempts for HSI classification have been made with deep learning [22], [23], in which convolutional neural network (CNN) has achieved many successes [24], [25]. Generally, CNN can conduct feature extraction over HSIs in different dimensions for classification tasks. For example, 1-dimensional CNN (1DCNN) directly handles 1-D spectral vectors to the network for classification [26], [27], by which existing relationships between the spectral signatures associated with each HSI pixel and the information contained in them are learned [28], [29]. In order to learn spatial features representations from the data, 2-D-CNN is used to handle dimensionality-reduced hyperspectral data by PCA for classification [30]–[32]. In order to well explore the high-dimensional data structure of HSIs, 3-D convolution is directly used in many CNNs to explore the spatial–spectral property of HSIs for classification [33], [34], such as multiscale 3D deep CNN (M3D-CNN) [35] and HSI-CNN [36]. The lightweight version of CNN-based HSI classification has also been explored in [37] and [38]. The autoencoders (AEs) have also been used as deep models to perform unsupervised coding from HSI data. For example, an unsupervised tied AE (TAE) was proposed for spectral feature extraction [39]. Spectral–spatial feature extraction has also been implemented using AE-based networks, such as stacked AE

Manuscript received February 7, 2021; revised June 24, 2021 and July 16, 2021; accepted July 27, 2021. Date of publication August 11, 2021; date of current version January 17, 2022. This work was supported in part by the Fundamental Research Funds for the Central Universities and in part by the National Natural Science Foundation of China under Grant 61671383. *(Corresponding author: Shaohui Mei.)*

Shaohui Mei, Xingang Li, and Xiao Liu are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, Shaanxi 710129, China (e-mail: meish@nwpu.edu.cn).

Huimin Cai is with Tianjin Jinhang Institute of Technical Physics, Tianjin 300192, China.

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA.

Digital Object Identifier 10.1109/TGRS.2021.3102034

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(SAE) [28], spatial updated deep autoencoder (SDAE) [40], two stacked sparse AEs (SSAEs) [41], compact and discriminative SAE (CDSAE) [42], and 3-D convolutional AE (3-D-CAE) [43].

While CNN-based methods have made significant progress [44], [45], other deep learning-based methods also attempt to classify HSIs, such as recurrent neural network (RNN), long short-time memory (LSTM) [46], and gated recurrent unit (GRU) [47]. For example, an RNN-based hyperspectral classification framework was proposed by treating hyperspectral pixels as sequential data and determining information categories via network reasoning [48]. A local spatial sequential method used in an RNN was also proposed, which can extract local and semantic information [49]. RNNs and CNNs have also been combined as convolutional RNN (CRNN) in [50], in which middle-level and locally invariant features learned in a few convolutional layers were then classified by recurrent layers. The spectral LSTM classification has also been combined with PCA extracted spatial LSTM classification via decision fusion [51]. The 2-D-CRNN and 3-D-CRNN have also been proposed for HSI data classification to perform a direct comparison with their CNN counterparts and demonstrating the superiority of the proposed RNN models [52]. Generally, these RNN-based methods take sequential data as input, recurse in the direction of evolution of the sequence, and connect all recurrent units in a chain. RNN regards an HSI as sequential data along the spectral direction and focuses on learning the HSI spectral sequence to predict the pixel category [48], [49]. Although the HSI classification architecture based on RNN has achieved good results, some problems are still unsolved. First, existing RNN-based works mainly treat each pixel an HSI as ordered sequence along the spectral direction, which cannot fully explore spectral similarity within an HSI. In addition, the spatial context of pixels, which is of crucial importance in classification tasks, cannot be directly used in conventional RNNs. Though spatial–spectral feature learning has been conducted using bidirectional-Convolutional LSTM for classification of HSIs [53], the spatial feature and spectral feature are not well explored since they are explored separately and concatenated for classification.

In view of the above problems, we use RNNs to fully explore the bidirectional relation between the adjacent band images within an HSI. In this article, a bidirectional long short-term memory (Bi-LSTM)-based network is designed for HSI classification. Specifically, for each pixel of HSI, we take its neighborhood block as the input of the attention network and pass through position attention network and channel attention network successively. These networks simultaneously utilize spatial and spectral information of a target pixel, which emphasizes the effective information and reduces the redundant information among spatial–spectral context of pixels. The resulted attention-based spatial–spectral vector is then fed into the Bi-LSTM for sequence to obtain the spatial–spectral feature of a target pixel, which is finally input to a softmax classifier to obtain the classification result.

In summary, the main contributions of this article are two-folded.

1) The bidirectional correlation of band images is considered using a Bi-LSTM network, in which both forward and backward correlation are considered simultaneously for classification of HSIs.
2) A spatial–spectral attention mechanism is designed in the proposed Bi-LSTM network to emphasize the effective information and weaken the redundant information among spatial–spectral context of pixels, by which the spatial and spectral contexts of pixels are weighted to alleviate spectral variation.

The remainder of this article is organized as follows. Section II introduces the proposed Bi-LSTM-based HSI classification strategy. The descriptions of datasets and experimental results are presented in Section III. Finally, Section IV concludes this article.

II. Proposed Method

A. Architecture of the Bi-LSTM

As shown in Fig. 1, in the proposed Bi-LSTM-based HSI classification strategy, the original hyperspectral data are first preprocessed using blocking and normalization and then fed to a spatial–spectral attention module and a Bi-LSTM prediction module sequentially. In the attention module, the spatial–spectral information is fused, by which the spatial and spectral context of pixels are weighted using the attention mechanism to alleviate spectral variation. Meanwhile, it allows the network to handle the 3-D input, which establishes a bridge between the conventional LSTM that only supports 2-D data input and the HSI data with three dimensions. In the Bi-LSTM prediction module, feature extraction is conducted by exploring both forward and backward correlation along spectral dimension simultaneously. Finally, the extracted spatial–spectral joint features are classified using a classic softmax classifier.

B. Spatial–Spectral Attention Module

Though HSI is well known for its high spectral resolution, spatial context information also plays an important role. However, the 3-D information of HSIs cannot be directly handled in conventional RNNs. In addition, since HSIs generally own a large number of spectral bands, many recurrent units are required to process spectral information, leading to the problem of vanishing gradient. Therefore, in this article, an attention mechanism for spatial and spectral information extraction is proposed, by which the 3-D information of HSIs can be reduced to meet the input requirement of RNNs.

Generally, the attention mechanism can improve the expressive ability of the network by emphasizing dominated information and suppressing insignificant information of the input. In this article, a spatial–spectral attention module is designed and conducted sequentially to selectively emphasize significant spatial context and spectral information of HSIs and thereby avoiding gradient disappearance or explosion. Specifically, as shown in Fig. 2, in terms of spatial context of HSIs, a position attention network is designed to selectively extract features that are conducive to target location category determination. On the other hand, in terms of spectral band

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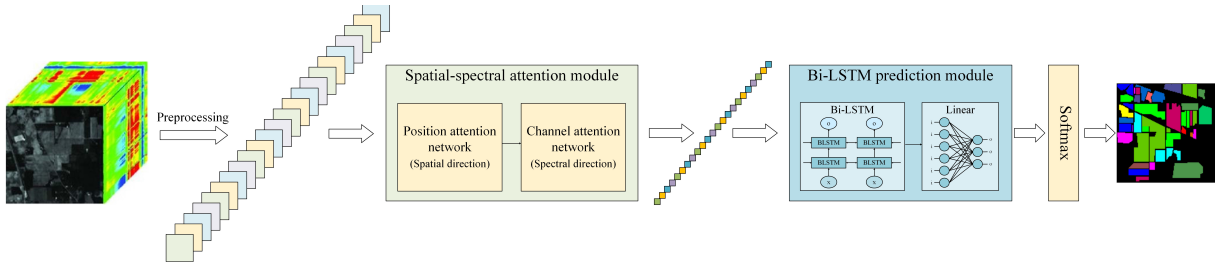


Fig. 1. Flowchart of the proposed Bi-LSTM strategy for hyperspectral classification.

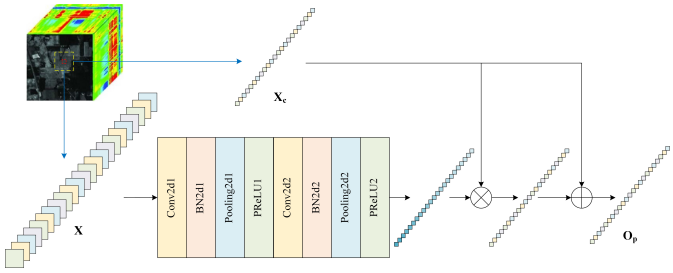


Fig. 2. Illustration of the position attention network.

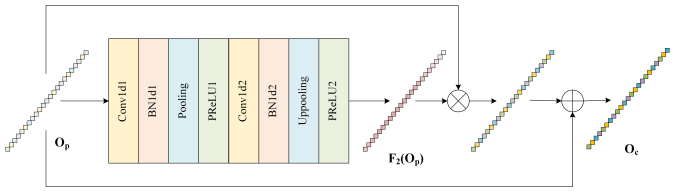


Fig. 3. Illustration of the channel attention network.

relationships, as shown in Fig. 3, a channel attention network is designed to selectively emphasize the spectral band information with prominent features and at the same time suppress the bands with relatively insignificant features.

Generally, the attention mechanism is modeled as follows:

$$\mathbf{O} = \mathbf{X} + \mathbf{X} \otimes \mathbf{F}(\mathbf{X}) \tag{1}$$

where $\mathbf{X}$ is the input, $\mathbf{F}(\cdot)$ represents the weight in a designed attention model, "$\otimes$" represents element-wise product, and $\mathbf{O}$ is the output. In the attention mechanism $\mathbf{F}(\cdot)$, the weighted summation of its input is conducted to highlight the key points and reduce redundancy. In order to classify pixel $\mathbf{X_c} \in R^{1\times1\times C}$, its spatially neighboring pixels, denoted as $\mathbf{X} \in R^{H\times W\times C}$, is considered as input to the network for spatial–spectral feature extraction, where H, W, and C are the length, width, and number of bands, respectively. As shown in Fig. 1, input data are first fed into the position attention model owning two convolutional blocks in the spatial direction, which is formulated as

$$\mathbf{O_p} = \mathbf{X_c} + \mathbf{X_c} \otimes \mathbf{F_p}(\mathbf{X}). \tag{2}$$

Here, $\mathbf{O_p} \in 1 \times 1 \times C$ is the result of the position attention model and the weight in this spatial attention mechanism $\mathbf{F_p}(\cdot)$ is obtained as

$$\mathbf{F_p}(\cdot) = \delta\left(\mathbf{f}_{\mathrm{CBP_{2d}}}\left(\delta\left(\mathbf{f}_{\mathrm{CBP_{2d}(X)}}\right)\right)\right) \tag{3}$$

where $\mathbf{f}_{\mathrm{CBP_{2d}}}(\cdot)$ represents a set of network processing by a 2-D convolutional layer, a 2-D batch normalization layer, and a 2-D pooling layer, and $\delta(\cdot)$ means the Parametric Rectified Linear Unit (PRelu) activation function.

TABLE I

PARAMETERS IN THE POSITION ATTENTION NETWORK

	Layer	Kernel size	Stride	Padding	Feature map
f1	Conv2d1	3×3	2	1	$H \times W \times 2C$
	BN2d1	/	/	/	$H \times W \times 2C$
	Pooling2d1	/	2	/	$H/2 \times W/2 \times 2C$
	PeLU1	/	/	/	$H/2 \times W/2 \times 2C$
f2	Conv2d2	3×3	1	1	$H/2 \times W/2 \times C$
	BN2d2	/	/	/	$H/2 \times W/2 \times C$
	Pooling2d2	/	2	/	$1 \times 1 \times C$
	PeLU2	/	/	/	$1 \times 1 \times C$

TABLE II

PARAMETERS IN THE CHANNEL ATTENTION NETWORK

	Layer	Kernel size	Stride	Padding	Feature map
y1	Conv1d1	1×1	1	1	$1 \times 1 \times C$
	BN1d1	/	/	/	$1 \times 1 \times C$
	Pooling	/	2	/	$1 \times 1 \times C/2$
	PeLU1	/	/	/	$1 \times 1 \times C/2$
y2	Conv1d2	1×1	1	1	$1 \times 1 \times C/2$
	BN1d2	/	/	/	$1 \times 1 \times C/2$
	Uppooling	/	2	/	$1 \times 1 \times C$
	PeLU2	/	/	/	$1 \times 1 \times C$

The spatially emphasized data $\mathbf{O_p}$ are then inputted into an attention mechanism in the spectral direction through the channel attention model, which is formulated as

$$\mathbf{O_c} = \mathbf{O_p} + \mathbf{O_p} \otimes \mathbf{F_2}(\mathbf{O_p}). \tag{4}$$

Here, $\mathbf{O_c} \in 1 \times 1 \times C$ is the results of the spectral attention model, i.e., the output of the proposed spatial–spectral attention mechanism, and the weight in this spectral attention mechanism $\mathbf{F_c}(\cdot)$ is obtained as

$$\mathbf{F_2}(\cdot) = \delta\left(\mathbf{f}_{\mathrm{CBU_{1d}}}\left(\delta\left(\mathbf{f}_{\mathrm{CBP_{1d}}}\right)\right)\right) \tag{5}$$

where $\mathbf{f}_{\mathrm{CBP_{1d}}}(\cdot)$ represents a set of network processing by a 1-D convolutional layer, a 1-D batch normalization layer, and a 1-D pooling layer.

In summary, the proposed spatial–spectral module conducts spatial attention and spectral attention sequentially. The parameters of the proposed position attention network and the channel attention network are shown in Tables I and II, respectively.

C. Bi-LSTM Prediction Module

As shown in Fig. 4, the proposed Bi-LSTM prediction module is composed of LSTM, connection layer, and linear

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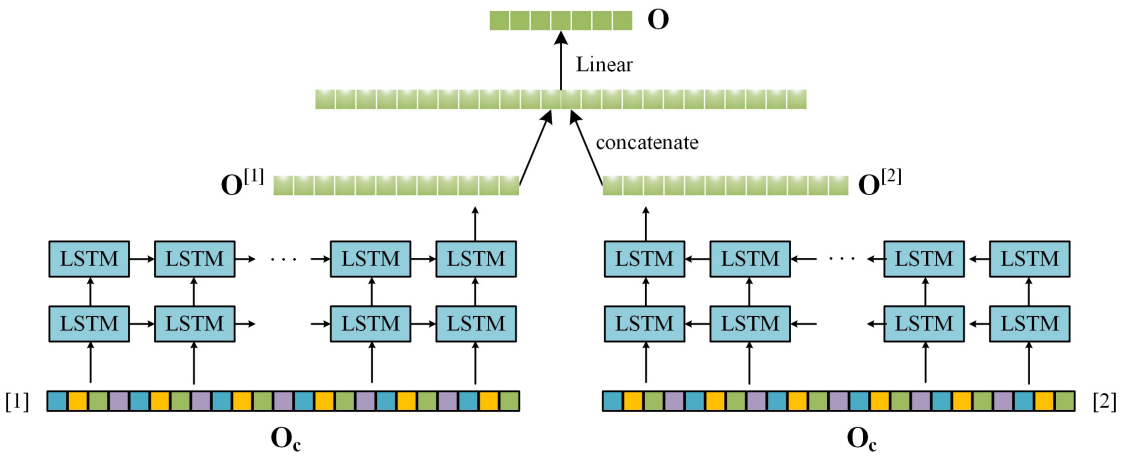


Fig. 4. Illustration of the proposed Bi-LSTM prediction module.

layer. Specifically, the vector produced by the spatial–spectral attention module, i.e., $\mathbf{O_c}$, is fed into several LSTM units in two directions: front-to-back ([1] in Fig. 4) and back-to-front ([2] in Fig. 4). Such two layers of LSTMs separately model the forward and backward correlation within the provided features $\mathbf{O_c}$ and generate two feature vectors: $\mathbf{O}^{[1]}$ and $\mathbf{O}^{[2]}$. These two feature vectors are further concatenated to a linear layer for feature integration. The integrated bidirectional features are finally fed to a softmax layer for classification. Therefore, the calculation process of Bi-LSTM prediction module can be expressed as

$$\mathbf{O} = \text{Linear}\big(\text{Con}\big(\mathbf{O}^{[1]}, \mathbf{O}^{[2]}\big)\big) \tag{6}$$

where Con$(\cdot)$ denotes connecting the bidirectional calculation results (matrix splicing) and Linear$(\cdot)$ represents the linear layer.

In order to distinguish between input and output, we set the output of the spatial–spectral feature extraction module to $\mathbf{X} \in R^{1\times 1\times C}$, i.e., $\mathbf{O_c}$ in (4), and the output after the Bi-LSTM prediction module is as follows:

$$\mathbf{O}^t = \varphi\big(\mathbf{O}^{(t)[1]} \oplus \mathbf{O}^{(t)[2]}\big) + c \tag{7}$$

$$\mathbf{O}^{(t)[1]} = \delta\big(U^{[1]}\mathbf{O}^{(t-1)[1]} + W^{[1]}\mathbf{X}^{(t)} + b^{[1]}\big) \tag{8}$$

$$\mathbf{O}^{(t)[2]} = \delta\big(U^{[2]}\mathbf{O}^{(t+1)[2]} + W^{[2]}\mathbf{X}^{(t)} + b^{[2]}\big) \tag{9}$$

where φ is the activation function of the hidden layer, δ is the activation function of the output layer, U and W are the weight coefficients, b is the bias coefficient, and c is the constant coefficient. $\mathbf{O}$ is the network output, [1] stands the forward RNN, [2] is the reverse RNN, $\oplus$ represents the matrix splicing, and $\mathbf{O}^{[1]}$ and $\mathbf{O}^{[2]}$ are the hidden layer output, given by the following formula:

$$\mathbf{s}^{(t)} = \mathbf{g}_f^{(t)}\mathbf{s}^{(t-1)} + \mathbf{g}_i^{(t)}\delta_{\mathbf{s}}\big(W\mathbf{h}^{(t-1)} + U\mathbf{X}^{(t)} + b\big) \tag{10}$$

$$\mathbf{h}^{(t)} = \mathbf{g}_o^{(t)}\delta_h\big(\mathbf{s}^{(t)}\big) \tag{11}$$

$$\mathbf{g}_i^{(t)} = \mathbf{F_s}\big(W_i\mathbf{h}^{(t-1)} + U_i\mathbf{X}^{(t)} + b_i\big) \tag{12}$$

$$\mathbf{g}_f^{(t)} = \mathbf{F_s}\big(W_f\mathbf{h}^{(t-1)} + U_f\mathbf{X}^{(t)} + b_f\big) \tag{13}$$

$$\mathbf{g}_o^{(t)} = \mathbf{F_s}\big(W_o\mathbf{h}]^{(t-1)} + U_o\mathbf{X}^{(t)} + b_o\big) \tag{14}$$

where the subscripts i, f, and o represent the input gate, forget gate, and output gate of the LSTM unit, respectively, $\mathbf{h}$ is the system state, δ_h and δ_s are the activation function of the system state and the hidden layer state, respectively, which is set as the "tanh" function, b is the bias coefficient, c is the constant coefficient, $\mathbf{g}$ is the gating unit updated with the time step, W and U are the weight coefficients, b is the bias, and $\mathbf{F_s}$ is the sigmoid function.

In summary, the parameters of the Bi-LSTM module are shown in Table III.

TABLE III
PARAMETERS OF THE PROPOSED BI-LSTM MODULE

	Part	Value
Bi-LSTM	Input size	1
	Hidden size	64
	Num layers	2
	Drop out	0.5
Linear	In feature	64
	Out feature	class num

The 'class num' means the number of categories of hyperspectral images.

III. EXPERIMENTAL RESULTS

In this section, a series of experiments over three benchmark HSI datasets are conducted to evaluate the classification performance of our proposed Bi-LSTM-based framework. Six state-of-the-art methods are selected as the baseline for comparison purposes.

A. Datasets

Three benchmark hyperspectral datasets are selected in this experiments, including Salinas Valley dataset, Pavia Centre dataset, and Pavia University dataset.

1) Salinas Valley Dataset: The Salinas Valley dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the agricultural area of the Salinas Valley in California. Its spatial resolution is 3.7 m per pixel. The dataset image coverage area with a pixel size of 512×217 is covering 224 spectral bands. Only 200 bands are selected in this experiment by removing the low SNR bands, water absorption, and atmospheric absorption bands. As shown in Fig. 5, this area mainly contains 16 categories of ground materials, including vegetables, soil, and vineyards. The ratio of training samples to testing samples is set as 1:9, according

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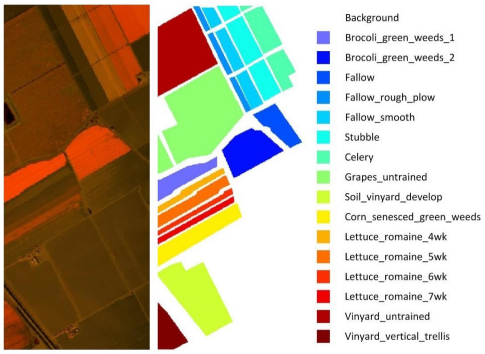


Fig. 5. (a) Pseudo-color image of the Salinas Valley dataset. (b) Ground-truth classification map for the Salinas Valley dataset.

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE SALINAS VALLEY DATASET

No.	Class	Total	Training	Testing
1	Brocoli_green_weeds_1	2009	201	1808
2	Brocoli_green_weeds_2	3726	373	3353
3	Fallow	1976	198	1778
4	Fallow_rough_plow	1394	140	1254
5	Fallow_smooth	2678	268	2410
6	Stubble	3959	396	3563
7	Celery	3579	358	3221
8	Grapes_untrained	11271	1128	10143
9	Soil_vinyard_develop	6203	621	5582
10	Corn_senesced_green_weeds	3278	328	2950
11	Lettuce_romaine_4wk	1068	107	961
12	Lettuce_romaine_5wk	1927	193	1734
13	Lettuce_romaine_6wk	916	92	824
14	Lettuce_romaine_7wk	1070	107	963
15	Vinyard_untrained	7268	727	6541
16	Vinyard_vertical_trellis	1807	181	1626

to which the number of samples in this experiment is listed in Table IV.

2) Pavia Centre Dataset: The Pavia Centre dataset was collected by the reflective optics system imaging spectrometer (ROSIS) in the center of Pavia in northern Italy. The dataset image is a 1096×715 pixel image over 102 spectral bands, with a spatial resolution of 1.3 m. This area mainly contains nine types of features, whose visual maps are shown in Fig. 6. By selecting 10% of labeled samples as training sample, the number of samples and the experimental settings for the training set and test set are shown in Table V.

3) Pavia University Dataset: The Pavia University dataset is obtained by sampling the vicinity of the University of Pavia in Italy using ROSIS sensors. The ground sampling interval of the sensor is 1.3 m, and the collected image contains 610×340 pixels and 103 spectral bands. There are nine types of ground data. The visual map of the feature category is shown in Fig. 7. By selecting 10% of labeled samples as training sample, the number of samples and the experimental settings for the training set and test set are shown in Table VI.

B. Experiment Setup

In the proposed Bi-LSTM, 9×9 spatial neighborhood is cropped as an input to the network for all of the three considered datasets. The number of experimental epochs is set to 500. The learning rate uses a learning rate decay

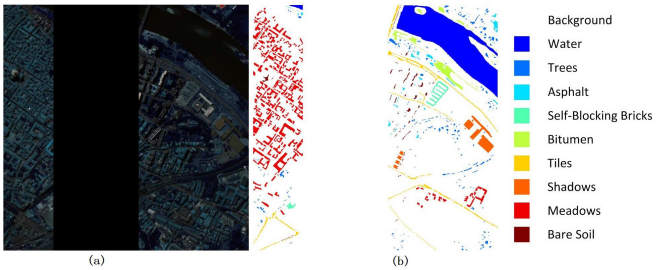


Fig. 6. (a) Pseudo-color image of the Pavia Centre dataset. (b) Ground-truth classification map of the Pavia Centre dataset.

TABLE V
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PAVIA CENTRE DATASET

No.	Class	Total	Training	Testing
1	Water	65971	6598	59373
2	Trees	7598	760	6838
3	Asphalt	3090	309	2781
4	Self-Blocking Bricks	2685	269	2416
5	Bitumen	6584	659	5925
6	Tiles	9248	925	8323
7	Shadows	7287	729	6558
8	Meadows	42826	4283	38543
9	Bare Soil	2863	287	2576

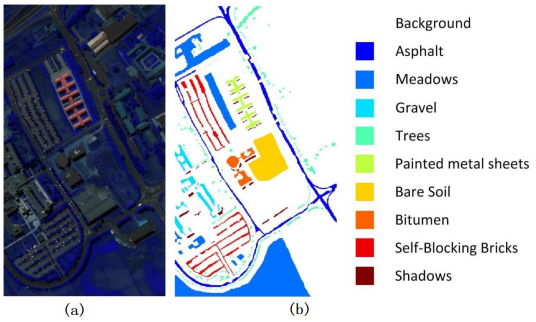


Fig. 7. (a) Pseudo-color image of the Pavia University dataset. (b) Ground-truth classification map of the Pavia University dataset.

TABLE VI
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Class	Total	Training	Testing
1	Asphalt	6631	664	5967
2	Meadows	18649	1865	16784
3	Gravel	2099	210	1889
4	Trees	3064	307	2757
5	Painted metal sheets	1345	135	1210
6	Bare Soil	5029	503	4526
7	Bitumen	1330	133	1197
8	Self-Blocking Bricks	3682	369	3313
9	Shadows	947	95	852

strategy, which is 0.0005 in the first 150 epochs and 0.00005 afterward, and the weight decay is set to 0.0005. NLLLoss, which is commonly used in multiclassification tasks, is selected as the loss function. In addition, the optimization method used in the experiment is Adam [54]. Several state-of-the-art deep learning-based HSI classification methods are selected for comparison, including RNN-based methods i.e., traditional RNN, LSTM [46], GRU [47], and deep RNN (DRNN) [48], and CNN-based methods, i.e., 1DCNN [26], CNN-based 3-D deep learning approach (3DDLA-CNN) [34], M3D-DCNN [35], and HybridSN [55]. Four evaluation metrics, including per-class accuracy, overall accuracy (OA),

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TABLE VII

QUANTITATIVE CLASSIFICATION RESULTS (%) USING DIFFERENT NETWORK-BASED METHODS OVER SALINAS VALLEY DATASET

No.	RNN	LSTM	GRU	DRNN	1DCNN	3DDLA-CNN	M3D-DCNN	HybridSN	Proposed method
1	91.24 ± 1.85	97.79 ± 0.30	97.78 ± 0.74	97.30 ± 0.14	98.62 ± 0.49	99.73 ± 0.28	99.80 ± 0.15	99.92 ± 0.08	99.87 ± 0.15
2	62.13 ± 7.21	97.59 ± 0.55	97.27 ± 1.02	99.06 ± 0.22	97.22 ± 1.99	99.57 ± 0.43	99.08 ± 0.42	99.21 ± 0.25	99.90 ± 0.08
3	58.79 ± 0.99	74.50 ± 5.50	66.34 ± 2.13	91.41 ± 3.13	93.67 ± 3.67	94.87 ± 4.98	95.55 ± 1.90	95.98 ± 2.62	99.67 ± 0.18
4	64.06 ± 2.55	96.79 ± 2.24	94.86 ± 1.95	98.35 ± 0.38	97.22 ± 4.23	99.58 ± 0.50	99.66 ± 0.38	100.00 ± 0.00	99.48 ± 0.27
5	88.71 ± 0.96	96.99 ± 1.28	95.42 ± 0.76	97.01 ± 0.71	95.22 ± 3.98	97.81 ± 0.80	97.51 ± 2.17	99.44 ± 0.23	99.39 ± 0.45
6	95.34 ± 0.23	98.51 ± 0.40	98.64 ± 0.28	99.89 ± 0.06	99.36 ± 0.39	99.74 ± 0.40	99.85 ± 0.16	99.71 ± 0.29	99.96 ± 0.12
7	52.22 ± 11.91	98.54 ± 0.50	97.71 ± 1.81	99.39 ± 0.09	97.77 ± 2.49	99.40 ± 0.41	99.89 ± 0.10	97.94 ± 2.06	99.38 ± 0.22
8	91.50 ± 2.60	78.65 ± 1.84	78.79 ± 1.89	80.49 ± 1.67	83.46 ± 2.46	84.54 ± 4.00	84.73 ± 3.61	95.70 ± 3.51	95.49 ± 0.97
9	97.69 ± 0.09	98.73 ± 0.13	98.08 ± 0.33	99.04 ± 0.25	98.32 ± 0.47	99.41 ± 0.18	99.46 ± 0.29	99.63 ± 0.10	99.91 ± 0.05
10	34.87 ± 3.73	74.38 ± 6.07	73.31 ± 6.23	85.08 ± 2.11	88.98 ± 4.68	92.06 ± 11.39	93.76 ± 0.79	96.53 ± 2.19	97.90 ± 0.40
11	0.00 ± 0.00	29.99 ± 23.00	46.25 ± 31.94	83.34 ± 3.82	93.35 ± 4.75	95.58 ± 9.24	98.79 ± 0.58	99.22 ± 0.05	97.63 ± 1.07
12	58.93 ± 2.80	87.27 ± 5.85	85.97 ± 3.51	93.73 ± 2.95	98.08 ± 2.07	98.50 ± 4.24	99.78 ± 0.33	99.08 ± 0.12	99.99 ± 0.02
13	45.78 ± 2.18	93.47 ± 5.57	93.63 ± 4.03	97.28 ± 0.35	97.58 ± 0.39	98.12 ± 3.26	98.33 ± 0.66	98.24 ± 1.76	99.79 ± 0.20
14	29.57 ± 5.73	84.34 ± 7.38	78.44 ± 6.24	93.05 ± 0.58	92.77 ± 1.78	97.72 ± 2.33	95.69 ± 1.33	96.37 ± 1.45	99.22 ± 0.69
15	3.20 ± 3.48	42.65 ± 7.73	36.60 ± 7.02	60.04 ± 2.53	52.24 ± 12.92	68.23 ± 7.64	72.25 ± 4.77	94.99 ± 4.66	95.62 ± 1.14
16	60.39 ± 11.89	93.11 ± 2.27	92.24 ± 3.56	96.34 ± 0.43	93.51 ± 5.68	95.67 ± 5.23	94.53 ± 1.40	99.23 ± 0.03	98.89 ± 0.38
AA	58.40 ± 2.33	83.96 ± 3.53	83.21 ± 2.55	91.93 ± 0.90	92.34 ± 2.26	95.03 ± 2.99	95.54 ± 0.45	98.20 ± 0.60	**98.88 ± 0.11**
OA	64.54 ± 1.71	82.06 ± 2.41	80.77 ± 1.60	87.98 ± 0.81	87.71 ± 2.38	91.20 ± 2.57	91.94 ± 0.46	97.61 ± 1.54	**98.12 ± 0.18**
Kappa	60.08 ± 1.95	79.95 ± 2.71	78.51 ± 1.81	86.60 ± 0.91	86.28 ± 2.68	90.19 ± 2.88	91.02 ± 0.51	97.34 ± 1.71	**97.91 ± 0.20**

TABLE VIII

QUANTITATIVE CLASSIFICATION RESULTS (%) USING DIFFERENT METHODS OVER PAVIA CENTRE DATASET

No.	RNN	LSTM	GRU	DRNN	1DCNN	3DDLA-CNN	M3D-DCNN	HybridSN	Proposed method
1	98.75 ± 0.00	99.90 ± 0.03	99.91 ± 0.01	99.83 ± 0.15	99.98 ± 0.01	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
2	75.79 ± 0.14	91.28 ± 2.83	92.85 ± 0.41	99.40 ± 0.12	93.71 ± 0.49	98.56 ± 0.38	99.18 ± 0.18	98.16 ± 1.27	98.81 ± 0.32
3	0.00 ± 0.00	87.09 ± 4.20	79.39 ± 6.84	89.00 ± 1.59	83.96 ± 2.24	96.86 ± 0.93	97.63 ± 0.39	95.80 ± 0.92	97.93 ± 0.62
4	0.00 ± 0.00	49.08 ± 4.98	44.91 ± 4.46	96.69 ± 0.71	72.26 ± 4.28	99.25 ± 0.20	98.26 ± 0.68	92.52 ± 5.40	98.39 ± 0.56
5	0.00 ± 0.00	62.07 ± 25.44	40.92 ± 16.98	95.49 ± 1.27	95.38 ± 0.71	99.59 ± 0.09	99.40 ± 0.32	99.73 ± 0.30	99.58 ± 0.17
6	93.54 ± 0.77	93.64 ± 1.36	92.81 ± 2.32	65.57 ± 4.70	95.31 ± 0.52	99.55 ± 0.16	99.61 ± 0.17	95.39 ± 2.79	99.38 ± 0.19
7	0.93 ± 1.06	82.99 ± 2.36	81.99 ± 4.30	87.61 ± 2.29	92.51 ± 0.84	99.06 ± 0.18	98.25 ± 0.11	98.23 ± 1.84	98.90 ± 0.35
8	95.06 ± 0.06	98.37 ± 0.45	97.96 ± 0.44	93.79 ± 0.65	99.64 ± 0.03	99.91 ± 0.02	99.94 ± 0.02	99.91 ± 0.06	99.93 ± 0.02
9	74.39 ± 0.10	99.64 ± 0.45	99.42 ± 0.57	99.98 ± 0.01	99.14 ± 0.22	99.69 ± 0.14	99.98 ± 0.05	97.63 ± 0.94	99.90 ± 0.08
AA	48.72 ± 0.04	84.89 ± 3.71	81.13 ± 2.99	91.93 ± 0.92	92.43 ± 0.91	99.16 ± 0.14	99.14 ± 0.11	97.49 ± 1.45	**99.20 ± 0.07**
OA	82.66 ± 0.01	94.92 ± 1.45	93.60 ± 1.13	97.67 ± 0.27	97.85 ± 0.22	99.72 ± 0.04	99.72 ± 0.03	99.22 ± 0.46	**99.73 ± 0.02**
Kappa	74.79 ± 0.01	92.76 ± 2.09	90.86 ± 1.62	96.70 ± 0.38	96.95 ± 0.31	99.61 ± 0.06	99.61 ± 0.04	98.90 ± 0.65	**99.62 ± 0.03**

TABLE IX

QUANTITATIVE CLASSIFICATION RESULTS (%) USING DIFFERENT METHODS OVER PAVIA UNIVERSITY DATASET

No.	RNN	LSTM	GRU	DRNN	1DCNN	3DDLA-CNN	M3D-DCNN	HybridSN	Proposed method
1	79.91 ± 1.34	92.64 ± 0.46	91.95 ± 1.84	87.78 ± 0.82	87.11 ± 3.14	96.10 ± 0.90	96.85 ± 1.70	97.47 ± 0.70	98.47 ± 0.37
2	98.44 ± 0.63	97.55 ± 0.59	97.37 ± 0.82	98.08 ± 0.14	94.47 ± 0.69	98.07 ± 0.58	99.30 ± 0.28	99.64 ± 0.00	99.91 ± 0.06
3	0.00 ± 0.00	4.72 ± 2.09	5.54 ± 2.08	20.91 ± 0.26	65.22 ± 3.19	85.07 ± 2.50	77.57 ± 5.76	91.19 ± 1.35	92.95 ± 1.71
4	79.78 ± 2.41	69.69 ± 10.33	80.41 ± 1.33	74.25 ± 1.05	84.64 ± 2.24	97.21 ± 0.88	98.03 ± 0.48	98.95 ± 0.33	99.00 ± 0.28
5	96.28 ± 0.08	72.76 ± 0.84	75.67 ± 7.26	98.26 ± 0.17	99.79 ± 0.10	99.75 ± 0.17	99.93 ± 0.17	99.96 ± 0.04	99.87 ± 0.28
6	6.36 ± 1.61	13.25 ± 3.97	14.99 ± 7.63	14.76 ± 1.10	57.94 ± 8.00	89.37 ± 2.07	94.25 ± 2.58	99.96 ± 0.04	99.41 ± 0.23
7	0.00 ± 0.00	0.07 ± 0.10	2.88 ± 6.41	7.69 ± 2.51	72.46 ± 5.16	88.12 ± 3.50	81.94 ± 7.13	87.05 ± 6.68	92.76 ± 1.97
8	84.74 ± 2.37	91.38 ± 0.55	89.90 ± 2.68	85.68 ± 1.28	79.18 ± 4.09	88.50 ± 1.92	88.54 ± 5.51	97.13 ± 0.09	96.55 ± 1.09
9	94.37 ± 0.23	99.93 ± 0.14	99.89 ± 0.10	98.24 ± 0.23	96.62 ± 2.27	94.82 ± 2.33	98.85 ± 0.78	90.02 ± 4.11	99.72 ± 0.34
AA	59.99 ± 0.07	60.22 ± 1.30	62.07 ± 2.20	65.07 ± 0.23	81.94 ± 2.72	93.00 ± 1.28	92.81 ± 2.16	95.71 ± 0.09	**97.63 ± 0.30**
OA	74.17 ± 0.33	76.04 ± 0.99	76.92 ± 0.94	77.32 ± 0.08	85.11 ± 2.28	94.89 ± 0.92	95.71 ± 1.39	98.07 ± 0.04	**98.71 ± 0.12**
Kappa	63.87 ± 0.38	66.44 ± 1.50	67.85 ± 1.41	68.34 ± 0.08	79.96 ± 3.14	93.21 ± 1.22	94.30 ± 1.85	97.44 ± 0.05	**98.29 ± 0.17**

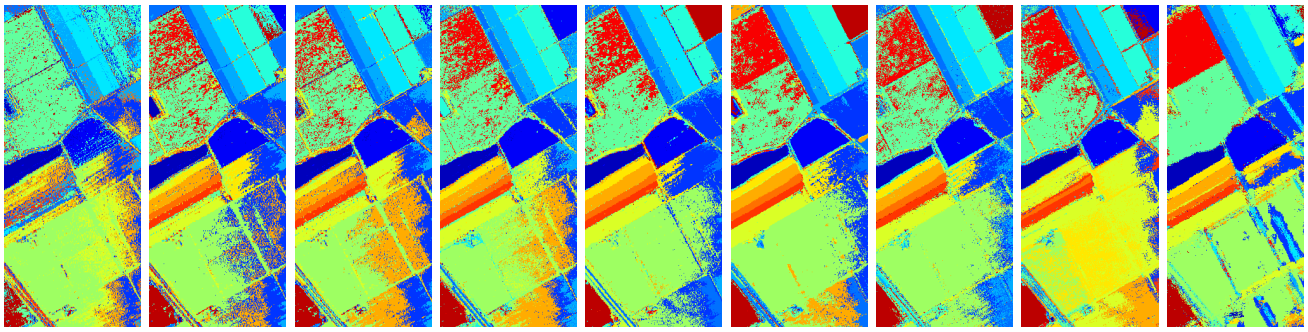
average accuracy (AA), and Kappa coefficient, are adopted for quantitative analysis. Note that tenfold cross validation is used for validation and both mean and standard variation of the results are reported.

C. Performance Analysis

The experimental results of the proposed Bi-LSTM over these three datasets, together with that of other state-of-the-art deep learning-based HSI classification methods, are reported in Tables VII–IX. It is observed that the Bi-LSTM obviously outperforms all of these considered methods over the three benchmark datasets.

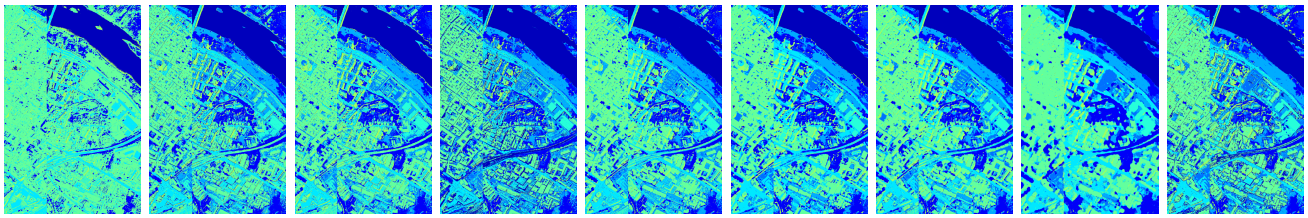
Among all the RNN-based classification methods, the traditional RNN performs the worst. This is due to the fact that the long-term dependence of RNN causes the redundant information of bands to greatly affect the network's prediction of pixel label. RNN cannot effectively process a large number of band data contained in HSIs. LSTM and GRU solve the problem of long-term dependence to a certain extent, by which their classification results have been greatly improved. However, since the spatial information is not considered, their performance is worse than other CNN-based algorithms that integrate the spatial–spectral information using 2-D and/or 3-D convolution. Among the CNN-based methods, 1DCNN is a ground-breaking classic HSI classification method. However,

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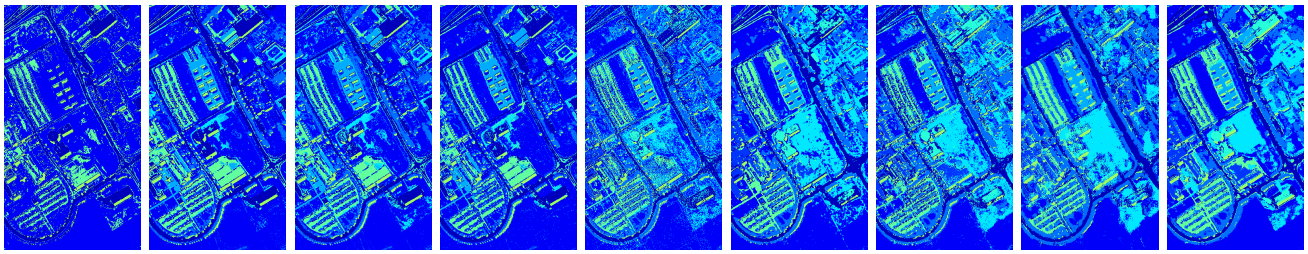
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 8. Classification maps of the Salinas Valley dataset. (a) RNN. (b) LSTM. (c) GRU. (d) DRNN. (e) 1DCNN. (f) 3DDLA-CNN. (g) M3D-DCNN. (h) HybirdSN. (i) Our proposed Bi-LSTM.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Classification maps of the Pavia Centre dataset. (a) RNN. (b) LSTM. (c) GRU. (d) DRNN. (e) 1DCNN. (f) 3DDLA-CNN. (g) M3D-DCNN. (h) HybirdSN. (i) Our proposed Bi-LSTM.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Classification maps of the Pavia University dataset. (a) RNN. (b) LSTM. (c) GRU. (d) DRNN. (e) 1DCNN. (f) 3DDLA-CNN. (g) M3D-DCNN. (h) HybirdSN. (i) Our proposed Bi-LSTM.

its performance is only better than RNN. This is because the spatial context is absent and its network structure is too simple. 3DDLA-CNN and M3D-DCNN explore both spatial context and spectral signature using 3-D convolution, by which their performances have been greatly improved. The HybirdSN further improves the performance by jointly using 2-D and 3-D convolution to explore the spatial–spectral feature. Compared with the RNN-based method, the proposed method combines the attention mechanism and explores the spatial–spectral joint information for HSI classification. Compared with the CNN-based method, the proposed method makes full use of the sequence characteristics of the HSI data. From a quantitative point of view, the proposed method has achieved superior performance over all three datasets, with the highest or the second highest accuracy in most categories, and has strong competitiveness. The classification results of comparison methods are shown in Figs. 8–10, which also indicates that the proposed Bi-LSTM provides the result that is closest to the ground truth.

The number of parameters, floating-point operations per second (FLOPs), and inference time of different algorithms over these three datasets are listed in Table X. Note that the inference time is measured with Intel Xeon Bronze 3104 CPU at 1.70 GHz with 125 GB of RAM with a single RTX2080Ti GPU card. It is observed that most of the RNN-based methods own a small number of parameters and FLOPs and less inference time. However, their performance is worse. The proposed Bi-LSTM owns more parameters and FLOPs than many CNN-based methods, which is only less than HybridSN. However, its performance obviously outperforms these CNN-based algorithms with a reasonable inference time.

D. Discussion

1) Impact of Attention Mechanism: The results of the proposed Bi-LSTM model with, without, and with partial of the designed attention mechanism are shown in Table XI. As observed from Table XI, the proposed position attention is more effective than channel attention in the proposed LSTM-based network structure, demonstrating that spatial context is very crucial to improve the performance of hyperspectral classification. The performance improvement by the proposed channel attention is not significant since the bidirectional spectral correlation has also been well explored in the proposed

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TABLE X
NUMBER OF PARAMETERS, FLOPS, AND INFERENCE TIME OF DIFFERENT ALGORITHMS

Dataset	Module	Parameters	FLOPs (K)	The execution time (s)
Salinas Valley	RNN	5,328	65	3.09
	LSTM	18,192	65	3.19
	GRU	13,904	65	3.24
	DRNN	247,888	17,547	3.26
	1DCNN	74,196	10,068	1.61
	3DDLA-CNN	45,196	283,809	2.12
	M3D-DCNN	273,120	1,444,391	2.51
	HybridSN	6,768,768	113,216,543	35.73
	Proposed method	1,635,720	1,394,120	11.70
Pavia Centre (Pavia University)	RNN	4,873(4,873)	36(36)	5.08(1.76)
	LSTM	17,737(17,737)	36(36)	5.02(1.68)
	GRU	13,449(13,449)	36(36)	5.02(1.79)
	DRNN	84,681(85,385)	5,849(5,906)	5.50(1.91)
	1DCNN	61,269(61,269)	5,295(5,310)	3.80(1.33)
	3DDLA-CNN	34,724(35,039)	140,204(143,063)	4.54(1.57)
	M3D-DCNN	81,881(81,881)	681,972(681,972)	5.39(1.83)
	HybridSN	4,887,801(4,906,233)	55,143,659(55,713,000)	47.36(14.30)
	Proposed method	510,267(517,656)	349,491(356,358)	17.72(5.51)

TABLE XI
CLASSIFICATION RESULTS (%) WITH DIFFERENT ATTENTION MECHANISMS

Dataset	Index	No attention	Position attention	Channel attention	Spatial-spectral attention
Salinas Valley	AA	92.45	98.61	92.49	**98.88**
	OA	87.96	97.39	88.24	**98.12**
	Kappa	86.58	97.09	86.87	**97.91**
Pavia Centre	AA	90.29	98.64	95.20	**99.20**
	OA	94.89	99.59	98.45	**99.73**
	Kappa	92.87	99.42	97.81	**99.62**
Pavia University	AA	86.37	97.60	87.22	**97.63**
	OA	87.96	98.65	89.74	**98.71**
	Kappa	83.78	98.22	86.20	**98.29**

Bi-LSTM structure. However, when joint using the proposed position attention and channel attention, the performance of the proposed LSTM algorithm can be clearly improved by more than 8% in all of the results, as listed in Table XI. The designed spatial–spectral attention mechanism can automatically emphasize both the spatial and the spectral information that are helpful for classification task in the training step. Therefore, the designed spatial–spectral attention mechanism is very effective to improve the classification performance under the proposed Bi-LSTM framework.

2) Impact of Unidirection and Bidirection: Table XII shows the effects of the proposed spatial–spectral attention mechanism with unidirectional and bidirectional models. It is observed that the bidirectional structure makes full use of the nondirectionality of the HSI sequence. Specifically, the OA of the bidirectional structure is more than 1% higher than that of the unidirectional structure on average, which fully reflects the superiority of the proposed bidirectional structure.

3) Impact of Spatial Size: In order to explore the effects of spatial sizes, different spatial sizes of cropped HSI cubes for these three datasets are considered, which is varied from 3×3 to 11×11. The classification results of the proposed Bi-LSTM method with different sizes of spatial inputs are reported in Table XIII. Moreover, the OA is further shown in Fig. 11 to discuss the impact of spatial size intuitively. It is observed that the classification performance first increases steadily and then decreases when the spatial size becomes larger. Generally, when a larger spatial size of cropped HSI cubes is fed to a network, more spatial information is considered for feature exploration and classification. However, the performance of classification does not always improve as the spatial size increases. As shown in Table XIII and Fig. 11, the classification performance begins to decline after spatial size of 9×9. This is mainly because more interference pixels that introduced by too large spatial size may affect the classification results.

TABLE XII
CLASSIFICATION RESULTS (%) OF THE PROPOSED SPATIAL–SPECTRAL ATTENTION MECHANISM WITH UNIDIRECTION AND BIDIRECTION

Dataset	Index	Unidirection	Bidirection
Salinas Valley	AA	97.94	**98.88**
	OA	96.76	**98.39**
	Kappa	97.55	**98.21**
Pavia Centre	AA	97.25	**98.89**
	OA	98.12	**99.65**
	Kappa	98.01	**99.50**
Pavia University	AA	97.28	**97.80**
	OA	97.24	**98.80**
	Kappa	96.77	**98.41**

4) Impact of Training Ratio: The spatial size of cropped HSI cubes is set to 9×9. The percentage of training samples varies from 2% to 10% with an increase of 2%. The classification of the proposed algorithm with a different number of training samples is reported in Table XIV, in which OA is also further shown in Fig. 12. It is observed that, in the case of scarce training samples (such as 2%), the network still has good performance, which shows that the network has a certain degree of robustness. With the increase of training samples, the performance of the proposed

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TABLE XIII

CLASSIFICATION RESULTS (%) WITH DIFFERENT SPATIAL SIZES

Dataset	Index	3×3	5×5	7×7	9×9	11×11
Salinas Valley	AA	95.83	97.30	98.03	**98.88**	98.74
	OA	93.92	94.88	95.84	**98.39**	96.90
	Kappa	93.22	94.30	95.27	**98.21**	96.55
Pavia Centre	AA	84.31	97.92	98.22	**98.89**	97.48
	OA	96.78	99.26	99.32	**99.65**	99.42
	Kappa	95.35	98.95	**99.54**	99.50	99.18
Pavia University	AA	92.77	94.73	96.14	**97.80**	97.58
	OA	94.96	96.43	97.50	**98.80**	98.65
	Kappa	93.30	95.27	96.70	**98.41**	98.22

TABLE XIV

CLASSIFICATION RESULTS (%) UNDER DIFFERENT RATIOS OF TRAINING SAMPLES

Dataset	Index	2%	4%	6%	8%	10%
Salinas Valley	AA	96.46	97.70	98.28	98.79	**98.88**
	OA	94.65	96.23	97.03	98.35	**98.39**
	Kappa	94.04	95.80	96.69	98.17	**98.21**
Pavia Centre	AA	95.31	95.74	98.14	**98.89**	98.81
	OA	98.69	98.85	99.26	99.56	**99.65**
	Kappa	98.14	98.37	98.95	99.38	**99.50**
Pavia University	AA	90.22	94.30	95.15	97.38	**97.80**
	OA	93.68	96.46	97.48	98.63	**98.80**
	Kappa	91.58	95.30	96.66	**98.41**	98.32

TABLE XV

CLASSIFICATION RESULTS (%) UNDER DIFFERENT METHODS ON THE DISJOINT INDIAN PINES DATASET

No.	LSTM	DRNN	1DCNN	3DDLA-CNN	M3D-DCNN	HybridSN	Proposed method
1	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	5.00 ± 6.32	3.50 ± 3.20	12.00 ± 17.49	11.00 ± 20.59
2	1.47 ± 1.35	28.63 ± 6.94	30.67 ± 23.12	38.58 ± 3.42	38.97 ± 8.18	23.22 ± 5.37	64.17 ± 6.88
3	0.23 ± 0.68	19.47 ± 6.33	13.40 ± 10.88	14.14 ± 2.66	7.27 ± 2.60	12.03 ± 5.77	21.35 ± 2.68
4	0.00 ± 0.00	2.86 ± 6.26	12.76 ± 15.61	28.00 ± 10.82	4.86 ± 4.01	3.62 ± 1.85	36.00 ± 11.78
5	0.00 ± 0.00	24.08 ± 8.10	6.90 ± 10.09	17.19 ± 2.74	6.80 ± 3.65	6.75 ± 5.38	27.15 ± 0.46
6	76.73 ± 5.95	90.20 ± 8.27	82.63 ± 5.48	75.47 ± 7.94	68.38 ± 8.52	37.54 ± 28.57	92.74 ± 4.64
7	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.71 ± 2.14	5.71 ± 11.43	30.71 ± 30.65
8	50.89 ± 18.10	100.00 ± 0.00	96.22 ± 5.24	72.86 ± 6.42	55.54 ± 10.17	91.07 ± 7.36	96.74 ± 5.81
9	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	9.00 ± 17.58
10	0.00 ± 0.00	34.62 ± 15.90	13.33 ± 18.66	15.38 ± 5.89	14.84 ± 5.65	23.82 ± 11.72	58.22 ± 12.76
11	97.23 ± 3.11	65.92 ± 6.10	75.24 ± 17.48	31.90 ± 4.66	40.25 ± 12.22	43.17 ± 8.94	44.91 ± 3.34
12	0.00 ± 0.00	3.11 ± 3.55	5.23 ± 5.74	11.19 ± 2.79	16.45 ± 5.30	32.29 ± 10.91	31.60 ± 17.81
13	0.00 ± 0.00	39.79 ± 29.97	50.07 ± 36.22	61.46 ± 4.42	30.52 ± 14.48	31.67 ± 29.69	80.21 ± 8.53
14	97.54 ± 1.04	92.27 ± 0.98	96.38 ± 2.61	79.32 ± 3.87	81.39 ± 7.94	74.91 ± 6.65	85.72 ± 6.46
15	0.79 ± 1.14	4.55 ± 5.44	8.29 ± 8.49	23.39 ± 5.54	26.24 ± 11.01	50.79 ± 27.45	42.91 ± 20.31
16	0.00 ± 0.00	45.45 ± 27.81	16.16 ± 27.63	16.97 ± 16.99	13.03 ± 12.86	1.82 ± 1.48	66.36 ± 16.17
AA	20.30 ± 1.12	34.43 ± 5.59	31.71 ± 7.16	30.68 ± 1.25	25.55 ± 2.59	28.15 ± 5.31	**49.93 ± 3.40**
OA	43.78 ± 0.62	49.98 ± 2.07	49.56 ± 2.83	38.76 ± 1.63	38.00 ± 1.79	38.07 ± 5.18	**56.62 ± 0.97**
Kappa	30.64 ± 1.07	41.42 ± 3.02	40.13 ± 4.88	31.07 ± 1.59	29.35 ± 2.14	29.38 ± 6.16	**50.85 ± 1.30**

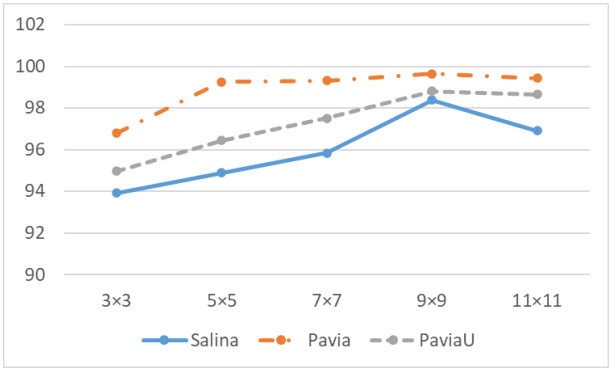


Fig. 11. OA(%) with different spatial sizes.

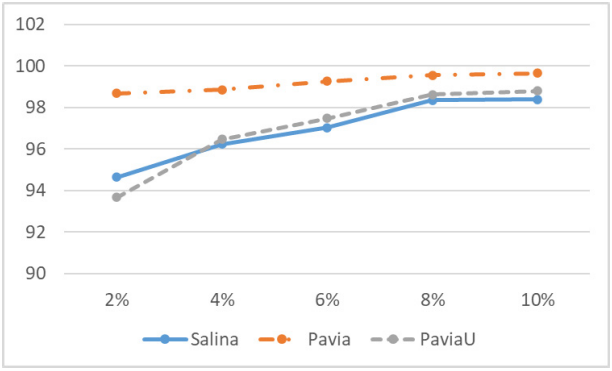


Fig. 12. OA(%) under different ratios of training samples.

method becomes better. However, the performance improvement becomes smaller when the number of training samples increases. This is because the training of the network will become steady when more and more training samples are used.

5) Experiment Over Disjoint Training–Testing Datasets: We further verify the performance of our proposed Bi-LSTM algorithm over the Indian Pines dataset using a disjoint training–testing strategy. As shown in Fig. 13(a), the Indian

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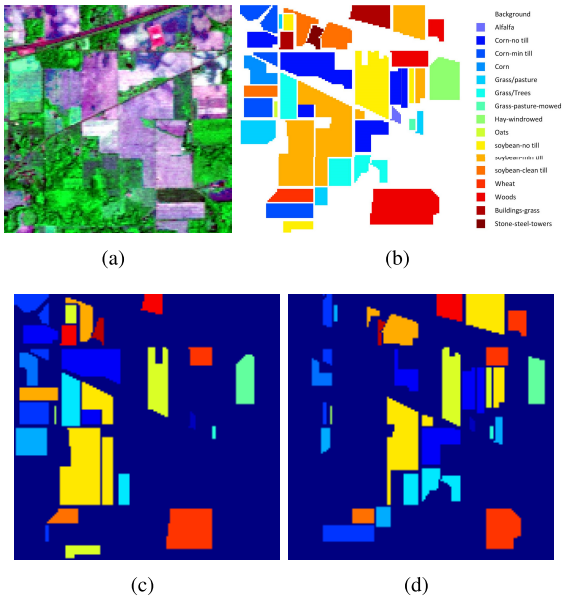


Fig. 13. (a) Pseudo-color image of the Indian Pine dataset. (b) Ground-truth classification map of Indian Pine dataset. (c) Ground-truth classification map of disjoint training samples. (d) Ground-truth classification map of disjoint testing samples.

TABLE XVI
EXPERIMENTAL RESULTS OF THE PROPOSED BI-LSTM WITH DIFFGRAD OR ADAM OPTIMIZERS

Dataset	Index	diffGrad	Adam
Salinas Valley	AA	97.39	98.88
	OA	95.29	98.12
	Kappa	94.75	97.91
Pavia Centre	AA	98.92	99.20
	OA	99.63	99.73
	Kappa	99.48	99.62
Pavia University	AA	97.39	97.63
	OA	98.60	98.71
	Kappa	98.15	98.29

Pine dataset contains 145 × 145 pixels with a ground resolution of 17 m. According to the ground-truth classification map of Indian Pine dataset shown in Fig. 13(b), 16 different land-cover classes of agriculture are mainly contained in this area. By using the disjoint training–testing strategy, the ground truth for training and testing is shown in Fig. 13(c) and (d), respectively. Note that only 20% of training samples (i.e., 10% of total labeled samples) in the training set are randomly selected for training and the average and standard variation of ten trails are used for evaluation. The quantitative results of the proposed algorithm, together with that of LSTM [46], DRNN [48], 1DCNN [26], 3DDLA-CNN [34], M3D-DCNN [35], and HybridSN [55], are shown in Table XV. It is observed that the performance of all these algorithms is very low when using such disjoint training–testing strategy. The networks only using spectral information (i.e., DRNN and 1DCNN) even outperform many CNNs integrating spatial and spectral information, such as 3DDLA-CNN, M3D-DCNN, and HybridSN. This is because limited spatial information is preserved after disjoint training–testing division of Indian Pine dataset, in which most pixels do not own enough spatial neighboring pixels that are required to be fed to networks in the testing image. However, our proposed attention-based Bi-LSTM clearly outperforms all the considered algorithms.

6) Experiment Using Different Optimization Strategies: We also train our proposed attention-based Bi-LSTM using other optimizers, such as diffGrad [56]. The comparison results of using the Adam optimizer versus diffGrad optimizer are shown in Table XVI. It is observed that, when replacing Adam by diffGrad, the performance of our proposed attention-based Bi-LSTM does not vary much over the two datasets acquired by ROSIS sensor, i.e., Pavia Centre and Pavia University datasets, while slightly decreased over the Salinas Valley dataset.

IV. CONCLUSION

In this article, a Bi-LSTM-based spatial–spectral HSI classification strategy is proposed by modeling the bidirectional correlation of hyperspectral pixels among adjacent bands. Moreover, a spatial and spectral attention mechanism is further used to handle spatial features, which cannot be processed by the conventional Bi-LSTM directly. Such a mechanism also highlights the important part of spatial and spectral information and reduces redundancy, by which the classification performance is further improved. Experimental results over three benchmark HSI datasets, including Salinas Valley dataset, Pavia Centre dataset, and Pavia University dataset, show that the proposed Bi-LSTM-based HSI classification strategy obviously outperforms several state-of-the-art deep learning-based strategies, including both RNN- and CNN-based ones. In future, we will try to explore various attention-based mechanisms for hyperspectral classification and directly implement attention-based mechanism into Bi-LSTM network. Moreover, large suitable hyperspectral datasets for disjoint training–testing validation are also considered.

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Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student with the University of Sydney, Camperdown, NSW, Australia, from October 2007 to October 2008. He is an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei was a recipient of the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014, the Best Paper Award of IEEE ISPACS 2017, and the Best Reviewer of IEEE JSTARS in 2019. He served as the Registration Chair for the IEEE China Summit and International Conference on Signal and Information Processing (ChinaSIP) 2014. He is serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) and a reviewer for more than 20 international famous academic journals.



Xingang Li received the B.Eng. degree from Dalian Maritime University, Dalian, China, in 2019. He is pursuing the master's degree in signal and information processing with Northwestern Polytechnical University, Xi'an, China.

His research interests include hyperspectral remote sensing image processing and deep learning.



Xiao Liu received the B.Eng. and the master's degrees in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2018 and 2021, respectively.

Her research interests include hyperspectral remote sensing image processing and deep learning.



Huimin Cai received the B.S. degree in transport engineering from Harbin Institute of Technology, Weihai, China, in 2007, and the Ph.D. degree in mechatronic engineering from Beijing Institute of Technology, Beijing, China, in 2012.

She is a Senior Engineer with Tianjin Jinhang Institute of Technical Physics, Tianjin, China. Her research interests include Infrared spectrum imaging detection technology and Infrared detection system design.

Dr. Cai serves as a Young Reviewer for the IEEE JOURNAL OF INFRARED AND LASER ENGINEERING. She made the conference report on the third and the fifth hyperspectral imaging technology and its application seminar in 2016 and 2020.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore, Baltimore, MD, USA, in 2000.

She is a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She served as the Co-Chair for the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013 and the Chair for the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She is the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image (HSI) and Signal Processing: Evolution in Remote Sensing in Shanghai, China, in 2012. She served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

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